EXPERIMENT 6: TIME SERIES ANALYSIS OF WEBSITE TRAFFIC

CODE :

```
import pandas as pd

import matplotlib.pyplot as plt

from statsmodels.tsa.seasonal import seasonal_decompose


# Load dataset

data = pd.read_csv(r'website_traffic.csv', parse_dates=['Date'],
index_col='Date')


# Time Series Decomposition

decomposition = seasonal_decompose(data['Page_Views'],
model='additive')

decomposition.plot()
```



```
plt.show()


# Moving Average

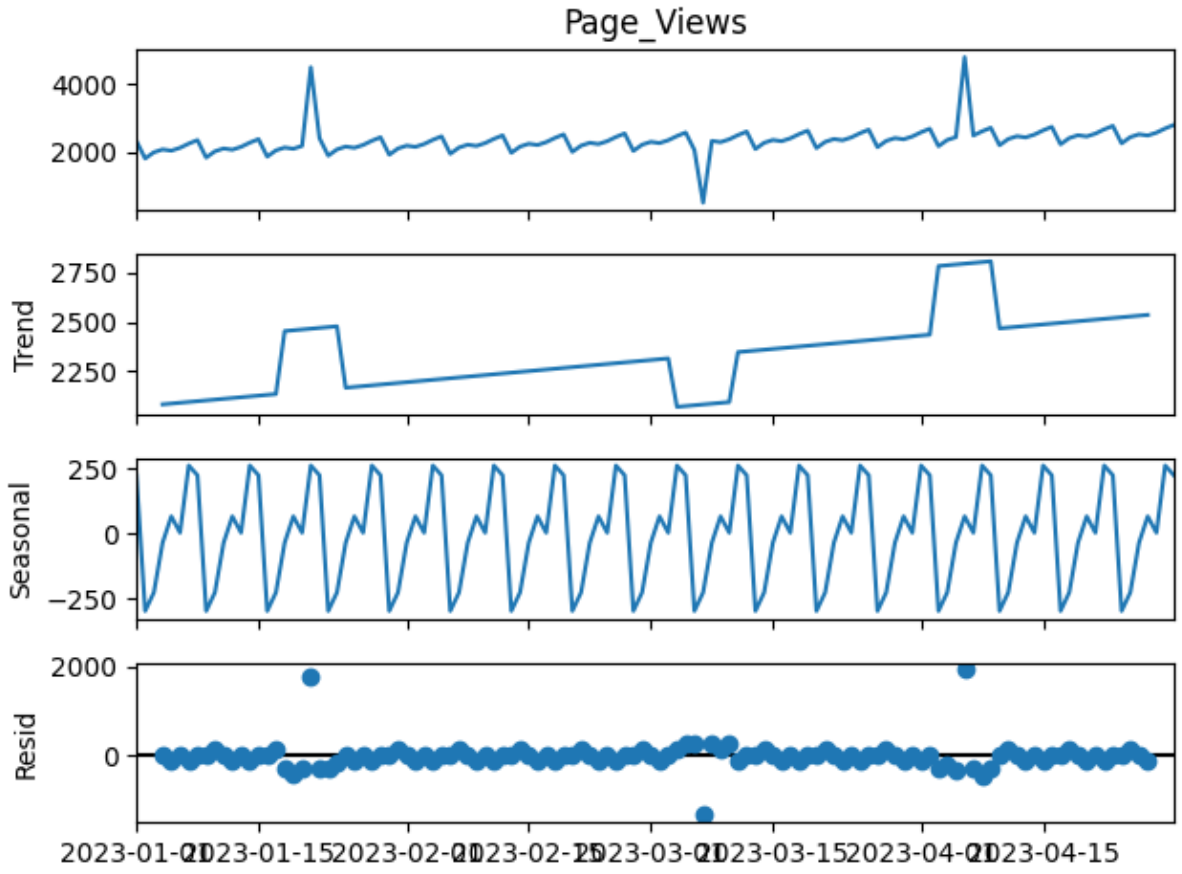
data['Page_Views_MA'] =
data['Page_Views'].rolling(window=30).mean()

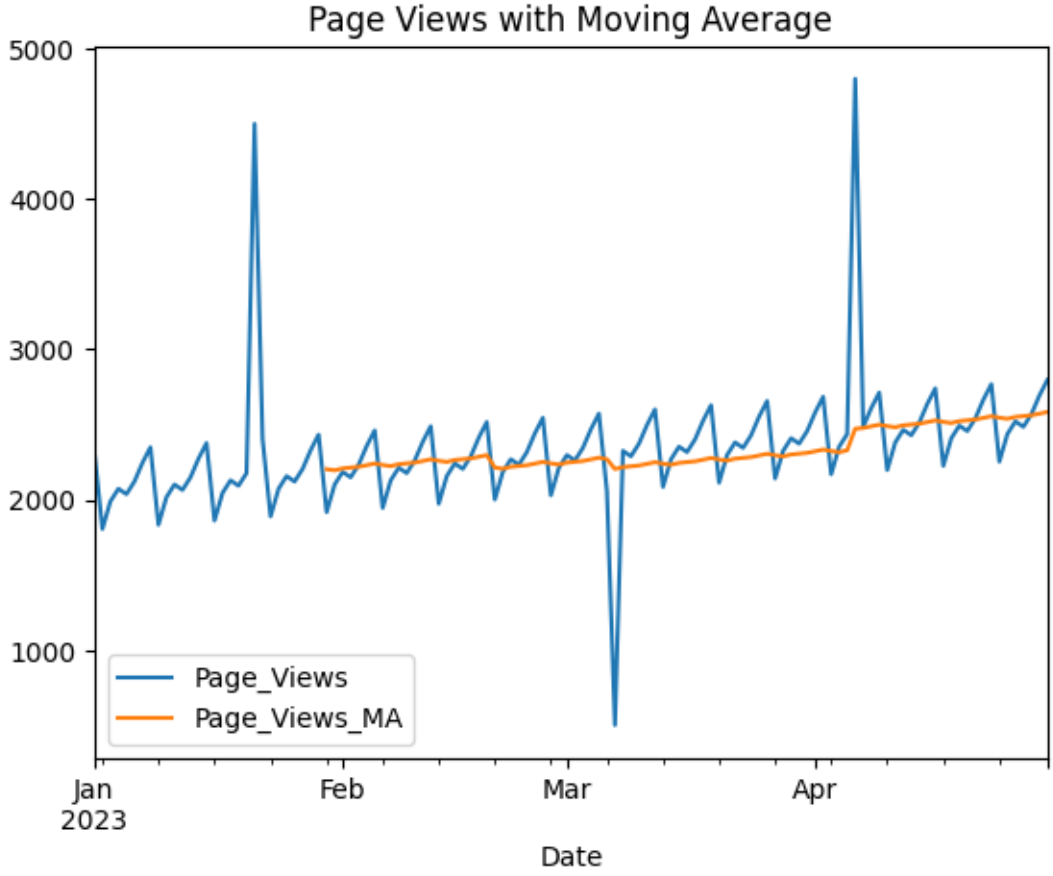
data[['Page_Views', 'Page_Views_MA']].plot()
```

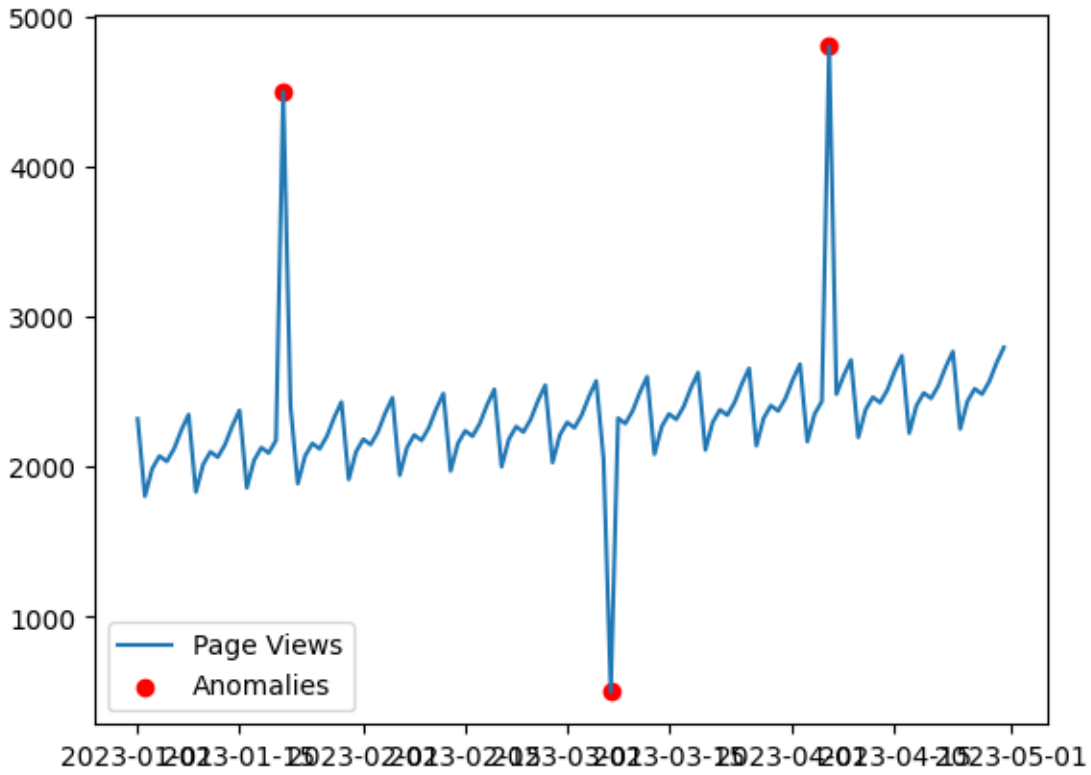
```
plt.title('Page Views with Moving Average')
plt.show()

# Anomaly Detection
data['Page_Views_Z'] = (data['Page_Views'] -
data['Page_Views'].mean()) /data['Page_Views'].std()
anomalies = data[data['Page_Views_Z'].abs() > 3]
plt.plot(data.index, data['Page_Views'], label='Page Views')
plt.scatter(anomalies.index, anomalies['Page_Views'], color='red',
label='Anomalies')
plt.legend()
plt.show()
```

OUTPUT :


Page_Views
4000
2000
Trend
2750
2500
2250
Seasonal
250
0
−250
Resid
2000
0
2023-01-01
2023-01-15
2023-02-01
2023-02-15
2023-03-01
2023-03-15
2023-04-01
2023-04-15


Page Views with Moving Average
5000
4000
3000
2000
1000
Page_Views
Page_Views_MA
Jan
2023
Feb
Mar
Apr
Date


5000
4000
3000
2000
1000
Page Views
Anomalies
2023-01-01
2023-01-15
2023-02-01
2023-02-15
2023-03-01
2023-03-15
2023-04-01
2023-04-15
2023-05-01